PLATFORM OVERVIEW

LogiYantra

AI-Powered Autonomous Logistics Intelligence

End-to-End Predictive & Autonomous Supply Chain Orchestration

7 Interconnected Machine Learning & Optimization Engines

Real-Time Telemetry Ingestion • Operational Decision Synthesis • Human Governance Gate

7 Modules
FULL-STACK AI

92.06%
DEMAND ACCURACY

0.36 Thresh
DELAY RECALL GATE

100% Fallback
OSM OFFLINE ENGINE

INDUSTRY CHALLENGE

The $1.7T Global Supply Chain Crisis

Why Modern Logistics Systems Fail Under Volatility

- Demand Volatility & Bullwhip Effect: Flawed inventory projections lead to either catastrophic stockouts or high carrying costs.
- Transit Blindspots & Carrier SLA Violations: Unforeseen transit delays cascade into factory shutdowns and missed customer commitments.
- Warehouse Bottlenecks & Demurrage Charges: Inefficient dock scheduling leads to yard gridlock and steep carrier detention fines.
- Siloed Operational Decision-Making: Route planners, warehouse managers, and procurement teams operate in isolated data silos.

$1.7 Trillion
GLOBAL DISRUPTION LOSS/YR

43%
SHIPMENTS EXPERIENCE DELAYS

3.5x Cost
EMERGENCY EXPEDITED FREIGHT

89%
MANAGERS LACK END-TO-END VISIBILITY

SOLUTION PARADIGM

The LogiYantra Paradigm Shift

From Reactive Firefighting to Autonomous Intelligence

- Continuous Multi-Model Forecasting: Deep sequence models (LSTM) and gradient boosting (XGBoost) predict issues hours and days in advance.
- Physical Road & Yard Graph Topology: OpenStreetMap road networks and live sensor telemetry feed real-time routing engines.
- Mathematical Policy Solvers: Dynamic stochastic safety stock and reorder point algorithms adapt continuously to demand shifts.
- Autonomous Synthesis with Human Governance: High-confidence directives are auto-dispatched, while critical interventions require operator sign-off.

Self-Predicting
CONTINUOUS ML MODELS

Self-Optimizing
DYNAMIC ROP & ROUTING

Transparent
SHAP EXPLAINABILITY

Human-Gated
ENTERPRISE GOVERNANCE

SYSTEM ARCHITECTURE

Unified 3-Tier Microservice Pipeline

Decoupled Data, Analytical Engine, and Autonomous Execution

Tier 1: Telemetry & Ingestion: Walmart Sales POS, DataCo Supply Chain, OSM Road Graph, IoT Yard Telemetry, TomTom Live Flow API.

Tier 2: Multi-Model ML & Optimization: Demand Regressor, Delay Classifier, Graph Optimizer, Yard Congestion LSTM, Stochastic Inventory Engine, Cross-Module Policy Synthesizer, SHAP Explainer.

Tier 3: Operator Interface & Action Dispatch: Multipage Glassmorphism Workbench, Interactive Leaflet Route Overlays, AI Logistics Chatbot, Automated Webhook Dispatch.

REST API
FLASK MICROSERVICES

Lazy Load
ZERO STARTUP SPIKE

Offline-First
DETERMINISTIC FALLBACKS

Glassmorphism
OPERATOR WORKBENCH

MODULE 1

Module 1: Demand Forecasting Engine

Multi-Horizon Sales Prediction via XGBoost & LSTM

Business Goal: Eliminate stockouts and reduce overstocking across distributed retail fulfillment centers.

Dataset & Benchmark: Walmart Weekly Sales benchmark dataset across 45 stores and 81 departments.

Feature Engineering (29 Features): 7/14/30-day lag features, rolling moving averages, CPI, fuel price, temperature index, holiday markdowns.

Model Performance: XGBoost Regressor achieves RMSE of 2,611.80 and Weighted Absolute Percentage Error (WAPE) of 7.94% (92.06% accuracy).

7.94%
WAPE ERROR RATE

29 Features
ENGINEERED SIGNALS

XGB + LSTM
ENSEMBLE FORECASTING

Daily/Weekly
DYNAMIC HORIZON

MODULE 2

Module 2: Delivery Delay Risk Prediction

High-Recall XGBoost Classifier on Global Transit Networks

Business Goal: Identify delayed shipments before they fail delivery SLAs, enabling proactive rerouting or carrier intervention.

Dataset: DataCo Global Supply Chain dataset encompassing 180,000+ orders across multi-modal shipping lines.

High-Dimensional Feature Space: 5,567 one-hot and numerical features spanning origin-destination ports, shipping modes, customer geography, and order traits.

Technical Breakthrough: Calibrated decision threshold at 0.36 prioritizing recall (ROC-AUC: 0.7453, PR-AUC: 0.8151) using custom byte-slice UBJSON artifact loading.

0.7453	0.8151
ROC-AUC	PR-AUC
0.36	**5,567**
CALIBRATED THRESHOLD	INPUT FEATURES

MODULE 3

Module 3: Dynamic Route & Traffic Optimization

OSMnx Graph Physics with TomTom Live Traffic Integration

Business Goal: Compute optimal multi-criteria transit paths minimizing travel time, fuel burn, and traffic exposure.

Road Graph Engine: OpenStreetMap road network graph parsed via OSMnx & NetworkX, calculating exact topological shortest paths.

Live Traffic & Incident Fusion: Real-time TomTom Flow & Incident APIs with automated fallback to OSM Free-Flow physical speed tables when API keys are absent.

Candidate Route Synthesis: Compares multiple Pareto-optimal routes (Fastest, Shortest, Balanced) with live distance, ETA, and congestion heatmap overlays.

OSMnx
TOPOLOGICAL GRAPH

Dijkstra & A*
CANDIDATE ROUTING

Live API
TOMTOM FLOW & INCIDENTS

Zero Failure
STATIC FALLBACK ENGINE

MODULE 4

Module 4: Warehouse Congestion Forecasting

Deep Sequence Modeling for Yard & Dock Capacity Prediction

Business Goal: Forecast warehouse and loading dock gridlock 2 to 4 hours in advance to avert driver detention penalties.

Sequence Architecture: Bidirectional LSTM Recurrent Neural Network operating on a rolling 12-hour temporal sequence window.

8 Yard Telemetry Variables: Inbound truck count, active dock bay occupancy %, staging density, forklift utilization, worker headcount, throughput pallets/hr.

Model Performance: Test ROC-AUC of 0.837 and 76.6% accuracy, outputting congestion probabilities that trigger dock rescheduling.

0.837
ROC-AUC SCORE

76.6%
CLASSIFICATION ACCURACY

12 Timesteps
SEQUENCE MEMORY

2-4 Hours
LEAD-TIME WARNING

MODULE 5

Module 5: Autonomous Inventory Optimization

Stochastic Reorder Points, Safety Stock & EOQ Batching

Business Goal: Maintain 98%+ customer service levels while minimizing tied-up capital and warehouse storage space.

Dynamic Reorder Point (ROP): ROP = (Lead Time x Daily Demand) + Safety Stock, dynamically adjusting to Module 1 forecasts.

Stochastic Safety Stock (SS): Calculated using SS = Z x sigma x sqrt(L), capturing supplier lead-time variance and demand volatility.

Autonomous Batching & Status: Normalizes reorders to supplier master pack sizes; flags inventory health as HEALTHY, REPLENISH, or CRITICAL_REPLENISH.

Dynamic ROP
LEAD-TIME ADAPTIVE

98% SLA
Z = 2.05 FACTOR

Pack Sizing
SUPPLIER BATCH MATCH

3 Health States
TRI-COLOR RISK ENGINE

MODULE 6

Module 6: Autonomous Operational Decision Engine

Cross-Functional Policy Synthesis & Action Dispatch

Business Goal: Synthesize isolated predictions into prioritized, unambiguous operational directives.

Cross-Module Policy Matrix: Evaluates simultaneous inputs: Demand spikes (M1), Carrier delay risks (M2), Dock congestion (M4), Stock criticality (M5).

Discrete Action Triggers: Automatically generates directives: REROUTE_SHIPMENT, EXPEDITE_ORDER, RESCHEDULE_DOCK_SLOT, BUFFER_STOCK_SURGE.

Enterprise Governance Gate: Classifies actions by risk level; executes routine actions automatically while requiring human approval for high-cost overrides.

Unified Brain

4-MODULE SYNTHESIS

Ranked Actions

PRIORITY QUEUE

Human-in-Loop

SAFETY GOVERNANCE

Audit Trail

TRANSPARENT REASONS

MODULE 7

Module 7: Explainability & What-If Simulation

SHAP TreeExplainer & Interactive Counterfactual Engine

Business Goal: Provide complete transparency for AI recommendations and enable risk-free scenario testing for planners.

SHAP Feature Attributions: Utilizes Shapley Additive exPlanations to decompose individual model predictions into exact feature contributions.

Transparent Heuristic Risk Engine: Normalizes composite risk across lead time, congestion, and demand with explicit weighting.

Counterfactual What-If Sandbox: Allows operators to simulate disruptions (e.g. +30% demand surge, +4 days supplier delay) and view impact deltas.

SHAP Values
FEATURE ATTRIBUTION

Delta Analysis
BASELINE VS SCENARIO

Zero Black-Box
AUDITABLE DECISIONS

Instant Sandbox
REAL-TIME SIMULATION

USER EXPERIENCE

Operator Console & Interactive Visualizations

Glassmorphism Interface Designed for High-Pressure Logistics

- Unified Executive Dashboard: Live KPI cards, system health monitors, cross-module status badges, and rapid drill-down links.
- Interactive Route Workbench: Leaflet.js map with route candidate comparison, live distance/ETA metrics, and traffic flow markers.
- Simulation Playground: Interactive sliders for testing fuel price, lead time disruptions, and inventory policy sensitivity.
- Logistics AI Chatbot: Built-in conversational intelligence assistant answering operational queries and providing instant diagnostics.

Leaflet.js
INTERACTIVE ROAD MAPS

Dark Theme
ERGONOMIC GLASS UI

Live Chatbot
DIAGNOSTIC ASSISTANT

Responsive
MULTI-DEVICE SUPPORT

ENGINEERING EXCELLENCE

Backend Engineering & Production Readiness

Microservice Architecture, Lazy Loading & Test Coverage

- Shared Flask REST API: Modular services with clean JSON schemas, route validation, and decoupled service layers.
- Lazy Singleton Pattern: ML models (XGBoost, Keras LSTM) load on-demand, ensuring instantaneous API boot times and minimal RAM usage.
- Resilient Offline Mode: Every external dependency (TomTom API, online graph loaders) has a verified deterministic local fallback.
- Automated Test Harness: Comprehensive Pytest suite validating model inference, API endpoint contracts, and end-to-end multi-module pipelines.

<200ms API RESPONSE TIME	**Lazy Load** MEMORY-EFFICIENT
100% ENDPOINT SCHEMA VALIDATED	**Pytest Suite** AUTOMATED CI/CD

MEASURED RESULTS

Empirical Results & Business Impact

Validated Accuracy Metrics Across All 7 Modules

Demand Forecasting: 7.94% WAPE (92.06% accuracy) across multi-department retail fulfillment.

Delivery Delay Recall: 0.7453 ROC-AUC with calibrated 0.36 threshold, capturing 84%+ of delayed shipments.

Warehouse Congestion: 0.837 ROC-AUC predicting dock gridlock 2-4 hours ahead of yard saturation.

Estimated ROI: ~22% reduction in transit buffer times, ~18% inventory carrying cost savings, and 35% fewer demurrage fines.

92.06%
FORECAST PRECISION

84%+
DELAY RECALL RATE

22% Cut
TRANSIT BUFFER TIME

18% Save
INVENTORY CARRYING COST

FUTURE HORIZONS

Future Roadmap & Enterprise Scalability

From Autonomous Logistics to Global Intelligent Supply Networks

- Multi-Agent Reinforcement Learning (MARL): Decentralized multi-truck cooperative dispatch and real-time fleet coordination.
- Enterprise ERP & WMS Connectors: Native plug-ins for SAP S/4HANA, Oracle SCM Cloud, and Manhattan Associates Warehouse Management.
- Green Logistics & Carbon Optimization: Multi-objective routing optimizing simultaneously for fuel economy and scope-3 CO2 emissions.
- Edge AI Telematics: On-device model quantization (TensorFlow Lite) for in-cabin driver units operating with zero connectivity.

MARL
FLEET COORDINATION

SAP/Oracle
ENTERPRISE CONNECTORS

Green Routing
SCOPE-3 CO2 TRACKING

Edge TFLite
IN-CABIN AI

CONCLUSION & DEMO

LogiYantra: The Future of Logistics

Think Smarter. Move Faster.

- Unified Architecture: Seamlessly unites demand forecasting, transit routing, yard congestion, and inventory replenishment.
- Actionable Intelligence: Moves beyond passive charts to generate concrete, high-confidence operational directives.
- Transparent & Trustworthy: Backed by SHAP explanations, counterfactual simulations, and human-in-the-loop safety gates.
- Ready for Production: Live Flask REST API, responsive Glassmorphism workbench, and verified ML model pipelines.

Full-Stack
7 INTEGRATED MODULES

Production Ready
FLASK + LEAFLET UI

Autonomous
GATED ACTION ENGINE

Open for Q&A
LIVE DEMONSTRATION